EMS602: Simulation Tools for Engineering Analysis and Design, CFD Coursework

Flow in a stenosed aorta – influence of the mesh and the numerical method

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Abstract

The haemodynamics of blood flow in a stenosed aorta play a crucial role in understanding and identifying cardiovascular diseases and the effect of fluid flow dynamics. This report investigated the fluid flow separation using Computation Fluid Dynamics (CFD). Initial parameters included an inlet velocity of *0.12* ms^{-1} and an outlet velocity of *-0.09* ms^{-1}. Blood flow was modelled as an incompressible Newtonian fluid with a constant density of *1044* kgm^{-3} under steady state laminar flow.

The CFD simulations were carried out using SIMPLE algorithm to measure the Navier-stokes equation. Blood flow was modelled as an incompressible Newtonian fluid which was steady under laminar flow. 4 different mesh types were analysed to assess convergence and accuracy with the application of first and second order discretisation. The main findings showed the 'other' mesh type converged faster in comparison to the fine, medium and coarse mesh, making it more effective in displaying the flow phenomena in the stenosed aorta. Second order discretisation had a better performance which provided a more accurate solution despite being less stable and more expensive. This discretisation helped moderate the effects artificial viscosity had on the blood flow, improving the reliability of the results.

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1. Introduction

1.1 Background Literature and Justification for Study

Aortic stenosis is amongst one of the most common valve complications associated with disease. Aortic stenosis causes the opening of the aortic valve to become narrower and leads to disrupted flow of blood from the left ventricle into the aorta (American Heart Association, 2020). This disruption in blood flow leads to changes in the haemodynamics of blood flow including blood pressure, velocity, and shear stresses. This flow behaviour can be modelled and analysed using a Computational Fluid Dynamics (CFD) approach. This technique is fundamentally based on the principles of fluid mechanics, where partial equations which determine fluid flows are solved numerically and allow for predictions of fluid behaviour to be made (Pourkashanian et al., 2023).

This report aimed to investigate the existence or extent of a flow separation distal to a stenosis of the aorta. This was achieved via the use of the Simcenter Star-CCM+ software, where relevant data was collected and analysed via the use of Excel.

1.2 Flow Physics

The viscous and incompressible blood flowing through a stenosed aorta has significant effects on its haemodynamic behaviour. The blood is modelled as a Newtonian fluid, exhibiting viscous effects which alter distributions of shear stress predominantly near the vessel walls. As the blood nears the stenosis, velocity and pressure of flow are significantly impacted. Parabolic velocity profiles prior to the stenosis are altered due to pre-stenotic acceleration and post-stenotic turbulence is observed after the stenotic region (ECG and Echo Learning, n.d.). The increase in diameter after the stenosis causes adverse pressure gradients which can result in downstream flow separation, leading to vortices and turbulence.

Various mesh types were investigated including coarse, medium, fine and mixed meshes. These meshes were investigated at both first order and second order accuracy discretisation. First-order accurate discretisation is more stable and computationally efficient than second order accuracy but is not as precise as its second-order counterpart. The effects of mesh resolution were investigated via analysis of mesh and residual convergence. The monitoring of these can help to assess the stability and accuracy of the final solution. All these factors can together provide a comprehensive analysis of flow behaviour, focusing on flow separation, shear stress variations and influence of stenosis on haemodynamic behaviour.

1.3 Model Assumptions

The haemodynamics of blood flow are highly complex due to the interactions between all blood components. In this study, simplifications were made to the model to reduce complexity and increase computational efficiency. This included fluid, flow, and boundary conditions. The Navier-Stokes equation was used as the basis for this investigation as blood is assumed to be both incompressible and Newtonian (Damno et al., 2021). Blood is typically assumed to be a Newtonian fluid in large vessels like the aorta due to high shear rates where the non-Newtonian behaviour of blood becomes negligible (Gijsen, van de Vosse and Janssen, 1999). The density of blood remains constant during flow, leading to the assumption that the fluid is incompressible.

Blood flow was modelled as laminar, this is a suitable assumption to be made as the majority of blood flow occurring in healthy human vasculature is laminar. (Zhu, Seo and Mittal, 2018). Due to blood being in a liquid state under normal physiological conditions, the flow is modelled as liquid, pulsatility

as a result of the cardiac cycle was ignored and flow was assumed to be steady for simplification of the model. Constant density of the blood flow was also modelled as the fluid is incompressible. The homogeneous fluid nature of blood also leads to the modelling of segregated flow, simplifying the simulation and focusing on flow behaviour rather than multiphase interactions (Resolved Analytics, 2020). Appropriate boundary conditions were also applied to ensure realistic simulation of the flow environment, such as no-slip conditions at vessel walls, ensuring zero velocity relative to the walls. The inlet was set to have a boundary type of velocity inlet, ensuring a uniform velocity profile, and the pressure outlet boundary type ensures constant pressure.

1.4 Flow Parameters and Generic Solver Settings

Parameters for the model were chosen to simplify the simulation and are not exactly replicative or realistic *in vivo* conditions, such as the modelling assumptions that blood is Newtonian and incompressible. In the model used, blood flows into the aorta with a vessel diameter of *20 mm*, through the inlet at a velocity of *0.12* ms^{-1} and traverses through the curved section of the aorta until it reaches the top outlet with a diameter of *5 mm* [Table 2]. Here, some of the blood leaves the aorta, while the rest continues to flow through the straight section of the aorta until it reaches the stenosis. At the stenosed area, the diameter decreases to a minimum of *15 mm* [Table 2]. After passing through the stenosis, the blood leaves the aorta via either the left outlet, which has a diameter of *10 mm*, or the right outlet which has a diameter of *10 mm*. The density of the blood is set to *1044 kgm*$^{-3}$ and the Newtonian viscosity to *3.65* $\times$ *10*$^{-3}$ *Pa* $\cdot$ *sec* [Table 2]. The inlet was defined as having a y-component of *0.132* ms^{-1} for the velocity vector. The top outlet had a velocity magnitude of *-0.09* ms^{-1} [Table 2]. After applying appropriate boundary conditions, reports were created to monitor the wall shear stress on the bottom wall and the pressure drop from the inlet to the right outlet. Stopping criteria was set to an initial maximum step value of 1000 and this was adjusted accordingly based on convergence.

2. Methodology

This study aimed to investigate the haemodynamic of blood through a stenosed aorta using Computation Fluid Dynamics. As described in *section 1.3,* the flow of blood flow was simplified to be modelled as a Newtonian incompressible fluid, with laminar steady flow. The geometry modelled by a CAD software was modelled as a simplified aorta with an area of stenosis of *15 mm* in diameter [Fig 1]. This model includes key anatomical features to help computationally replicate blood flow under physiological conditions [Table 2]. The boundaries of this model include the inlet as the ascending aorta, the right and left outlet as the iliac arteries and the top outlet as the left subclavian arteries.

The simulation initially began with applying first-order discretisation due to its stability, which ensured for a smoother convergence, and subsequently was changed to a second-order discretisation. This provided a more accurate approximation for the flow variables, specifically in areas with a high flow complexity such as the area of stenosis in the descending aorta. In comparison to first order, second order discretisation is more expensive and less stable.

On the 4 different meshes [Coarse, fine, medium and ‘others’], mesh refinement was performed to assess the influence resolution had on the accuracy of the results. Reports were created to monitor the wall shear stress on the wall of the stenosed region, where the y-direction was specified as the direction vector. Additionally, the pressure drop was measured from the inlet to the right outlet to measure the resistance to blood flow as a significant drop in pressure can lead to cardiovascular adverse effects (Lumen Learning, 2012). In this study, the term ‘Solver’ alludes to the algorithm which is used to solve linear equations arising from discretisation. The ‘SIMPLE’ algorithm is used to solve incompressible flow. Due to the simplicity of the model, there are no changes to the default values.

To ensure that the simulation for each mesh reached a steady state within the bounds of an appropriate number of iterations, stopping criteria were defined based on the change in conservation equations. The maximum number of iterations used was set to *1000*, to avoid the case running for an infinite period. The residuals were monitored for the continuity equation and the x and y momentum, whist a minimum limit of 1.10^{-4} was set to ensure that the solver stops when the residual falls below the value.

The simulation solved the time evolution for the conservation equations, predicting the pressure and the velocity fields at each time step. This was to ensure the simulation was correctly run and reached a steady state. This began with defining the initial flow and velocity fields which were previously set as stated in *section 1.4.* For the initial solution, a guess was made which was used for the starting point. The initial solution was applied, and the simulation was run. As the solution advanced, the residuals for the 3 equations dropped which indicated that a steady state was being approached. The stopping criterion was used to determine when the solution fully converged, and the solver was set to stop after approximately *130* iterations.

After convergence, the solution was examined to visualize the pressure and velocity fields. Contour plots were generated with consistent scales and settings. Zoomed-in views of these plots allowed for the analysis of specific flow phenomena, with identical view settings applied across all simulations to facilitate comparison [Tables 4-12]. XY plots were then exported and combined into a single scatter graph [Figs 2-3] for a visual comparison of how mesh refinement and mesh order affected the results.

3. Results

3.1 Effect of mesh refinement and 2nd order accuracy, flow field

The velocity contour and vector maps [Tables 4-12] provided a detailed visualization of the velocity field, with zoomed-in views used to analyse specific flow phenomena. In the stenosed region [Table 4], velocity increased significantly in the central flow path, while the vessel walls experienced near-zero velocity, consistent with the boundary layer theory (S. Aliu et al 2020). This was confirmed by the velocity vector maps [Table 6], which demonstrated that regions of highest velocity occur centrally at the stenosis, specifically more toward the right side of the vessel. Mesh refinement enhanced the definition of high-velocity regions, showing smoother transitions and more developed flow patterns. Using second-order numerical solutions further increased velocity magnitudes and enlarged regions of high pressure, indicating convergence toward a more accurate solution for maximum velocity. In the small branch [Table 7], velocity increased significantly after the initial bend in the aortic arch, with higher flow along the outer wall of the bend, driven by centrifugal forces and flow inertia. Velocity vector maps [Table 9] confirm this, showing reduced velocity along the inner (left) wall [Fig 1] of the artery. The magnified view of the Subclavian artery inlet [Tables 10-12] shows that velocity was highest on the outer (right) wall [Fig 1] of the branch. With mesh refinement and second-order solutions, a focused high-velocity channel was observed [Table 12], entering the Subclavian artery on the inner (left) wall of the inlet and continuing toward the outer (right) wall [Fig 1] downstream.

The observed velocity changes correlated with corresponding pressure variations. Pressure in the stenosis region [Table 5] increased at the walls; interestingly, the right wall has higher pressure compared to the left wall. In post-stenosis regions [Table 5], pressure decreased toward pre-stenosis levels. Mesh refinement highlighted high-pressure regions, and second-order solutions revealed generally higher pressures, particularly on the right wall, corresponding to high velocities in the velocity field. This suggests that flow asymmetry, influenced by the aortic arch, leads to increased pressure and wall loading on the right. Both the small branch and Subclavian artery views showed

maximum pressure at the Subclavian inlet, especially on the right side. Mesh refinement and second-order solutions further clarified this trend. Higher pressures are observed on the outer wall of the small branch [Table 8], while lower pressures are seen on the inner wall, with a similar effect at the stenosis where the right side experiences slightly higher pressure than the left.

Overall, the velocity field demonstrates that flow predominantly follows the path of least resistance, with high velocities in straight or unobstructed regions and high pressures where flow is interrupted. At the Subclavian Artery inlet [Tables 10-12], high velocity coincides with increased pressure, as fluid dynamics principles predict that pressure rises where flow encounters vessel walls, adhering to the no-slip boundary condition. In the stenosis, high pressure is observed at the narrowed walls, while the centre of the vessel experiences relatively low pressure. The stenosis reduces the cross-sectional area, causing velocity to increase and pressure to decrease, consistent with Bernoulli's principle. This principle relates the pressure drop to the velocity increase in fluid flow through a stenosed segment, assuming ideal, steady flow with negligible losses.

3.2 Accuracy of Flow Separation

Flow separation was assessed using XY plots [Figs 2-3] of X-position versus Y-velocity across a slice positioned downstream of the stenosis. The plots revealed regions of negative velocity, indicating backflow, with the most negative values occurring slightly toward the right side of the vessel, at an average X-position of 0.0025 m to the right of the origin. The observed asymmetry in backflow regions could be attributed to changes in velocity and pressure resulting from the initial bend [Fig 1]. The increased velocity near the right side of the vessel wall is driven by the flow dynamics influenced by the curvature of the bend. The presence of negative velocity downstream of the stenosis strongly suggests flow separation, a phenomenon associated with adverse pressure gradients causing fluid to reverse direction. This flow behaviour is consistent with the recirculation zones often observed in post-stenotic regions (Katritsis, D. G et al. 2010).

The "other" type mesh demonstrated slightly faster convergence compared to the fine, medium, and coarse meshes, with the fine mesh achieving the highest backflow velocity after the "other" mesh. Additionally, second order meshes converged to higher backflow velocities (-0.2 ms^{-1}) [Fig 2] compared to first order meshes (-0.18 ms^{-1}) [Fig 3], indicating improved accuracy with higher-order numerical solutions. This highlights the critical role of second-order accuracy in capturing complex flow phenomena, such as flow separation and backflow, which may be underestimated with first-order schemes. The higher backflow velocity emphasizes the extent of flow separation, which is more accurately resolved using refined meshes and second-order solutions. The residual convergence graphs [Figs 4-11] demonstrate that convergence was achieved for all simulations. Residuals represent the error between successive solutions during iterations, reflecting how well the governing equations (continuity and momentum) are satisfied. Continuity residuals measure mass conservation error, while x- and y-momentum residuals track momentum conservation in respective directions. All residuals decrease over iterations, achieving convergence, suggesting a sufficiently accurate solution for the mesh type and order used.

3.3 Artificial Viscosity Considerations

During the analysis, signs of artificial viscosity were evident, particularly in the first-order simulations. In regions with sharp flow gradients, such as near the stenosis and downstream recirculation zones, the first-order results showed excessive smoothing of velocity and pressure profiles, as well as inaccurate gradients in some regions. This is indicative of numerical diffusion, or artificial viscosity,

which can occur with lower-order discretization schemes. The use of second-order schemes and refined meshes helped mitigate these effects, allowing for more accurate capture of sharp gradients and flow phenomena.

4. Summary

This report investigated the haemodynamics of blood flow in a stenosed aorta using CFD simulations via the Simcenter Star-CCM+ software to evaluate the effects of mesh refinement and numerical accuracy on the prediction of flow behaviour.

As per the discussion in Section 2, the most accurate solution observed among the calculations was the second-order fine mesh, as the corresponding results demonstrated the most precision in resolving the velocity gradients, pressure transitions, and flow separation distal to the stenosis as compared to all other configurations.

The sequence-fine mesh demonstrated better convergence and lower residuals, as it showed fewer inconsistencies in regions with high velocity and pressure gradients as compared to the ‘other' mesh, which was affected by suboptimal grid alignment and local mesh quality issues, indicating that it more effectively balanced accuracy and numerical stability as compared to the ‘other’ mesh. Also, coarser meshes failed to resolve finer details such as separation points and vortices downstream of the stenosis as compared to finer meshes which provided the most accurate results with well-defined flow separations and smoother velocity transitions. Pressure and velocity profiles were also more representative of theoretical expectations in the finer meshes.

Across all simulations, the first-order accuracy produced less precise results as compared to second-order accuracy, particularly in regions with high flow complexity. Artificial viscosity effects were also seen in first order simulations, which led to smoothed velocity gradients and underestimated flow separation. Second-order accuracy although more computationally expensive, showed more accurate prediction of flow separation distal to the stenosis, as vortices and sharper velocity gradients were observed in the results using this configuration.

While second order fine meshes provide the most accurate results, they are computationally expensive, and engineers must balance the need for precision with available resources and optimise mesh resolution based on the importance of specific flow features. These results prove that mesh refinement is essential for capturing flow behaviour in CFD, and that finer meshes combined with higher-order numerical methods allow for better resolution of gradients and complex flow behaviours.

These findings are proven to be reliable by the flow features such as separation and vortices being resolved in alignment with the Bernoulli’s principle and the Navier-Stokes equations, and these patterns were shown across multiple meshes and numerical schemes. Future recommendations to further improve the accuracy and reliability of this analysis could involve using a non-Newtonian blood model to provide a more accurate depiction of blood's shear-dependent viscosity, especially in smaller vessels or regions of low shear rates and further refinement of the mesh in areas of high gradients and validation against experimental or clinical data.

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6. Appendix

Table 1. Table showing all assumptions used in the model.

Model property/part	Assumption
Flow	Steady
	Laminar
	Segregated Flow
	Constant Density
	Liquid
Fluid	Newtonian
	Incompressible
Boundary Conditions	No-slip

Figure 1. Geometry of the model, labelled with the diameters listed in **Table 2**.

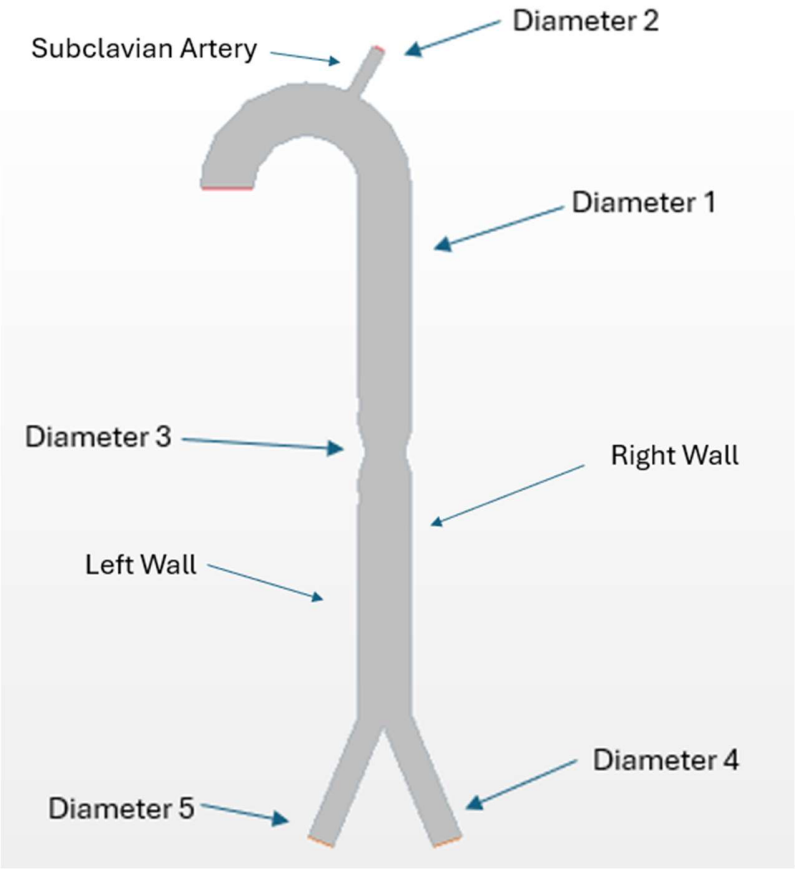


Table 2. Table showing dimensions and initial conditions for the model.

Property	Value
Initial Dimensions:	
Diameter of Main Vessel (1)	0.020 m
Diameter of Small branch (2)	0.005 m
Diameter of Stenosis (3)	0.015 m
Diameter of Right Outlet (4)	0.012 m
Diameter of Left Outlet (5)	0.010 m
Inflow velocity (Main vessel)	0.12 m/s
Outflow velocity (Small vessel)	0.09 m/s
Fluid Density	1044 kg/m^3
Fluid Viscosity	0.00365 Pa.S

Reference Values:	
Maximum allowable Absolute Pressure	$1*10^8$ Pa
Minimum allowable Absolute Pressure	1000 Pa
Reference Pressure	101325 Pa
Initial Conditions:	
Pressure	0 Pa
Velocity	0 m/s

Table 3. Table displaying Contour settings and Scale bars.

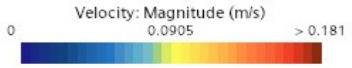
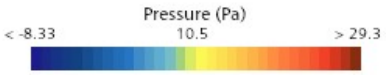
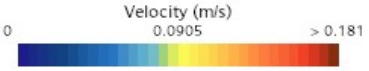
Property	**Value**
Velocity magnitude scale	
Pressure magnitude scale	
Velocity vector scale	
Number of levels in Colour Bar	32

Table 4. Contour plots of velocity magnitude in the stenosis region, comparing various mesh types with 1st- and 2nd-order discretization.

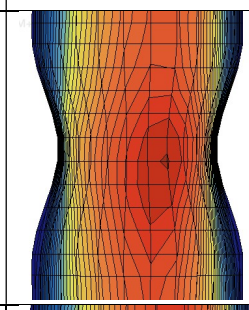
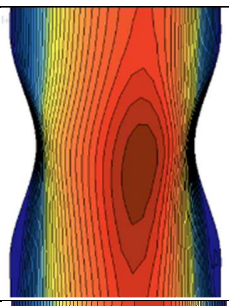
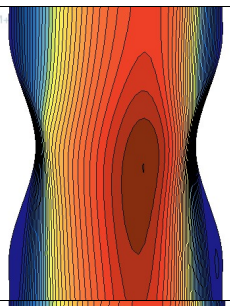
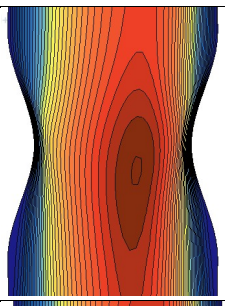
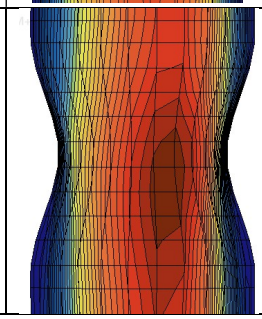
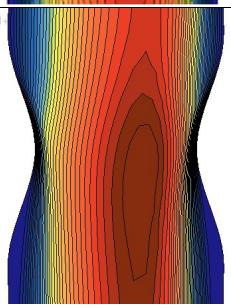
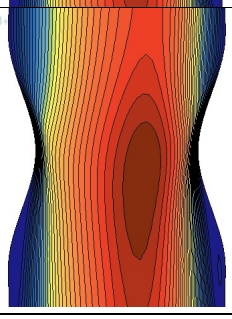
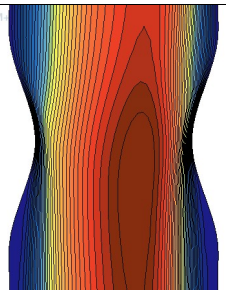
Type of mesh:	**Coarse**	**Medium**	**Fine**	**Other**
1st order				
2nd order				

Table 5. Contour plots of pressure magnitude in the stenosis region, comparing various mesh types with 1st- and 2nd-order discretization.

Type of mesh:	**Coarse**	**Medium**	**Fine**	**Other**

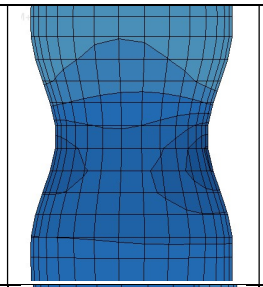
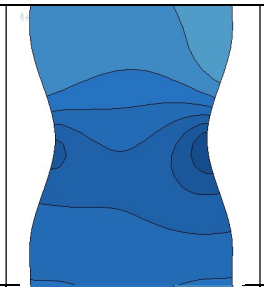
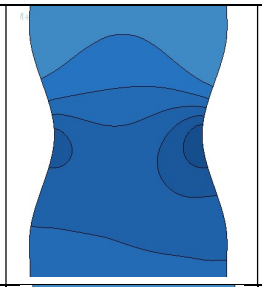
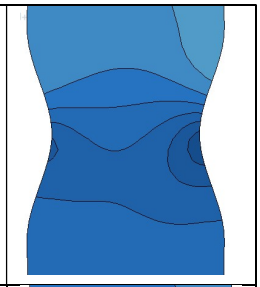
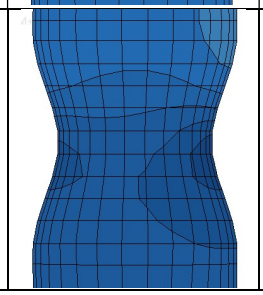
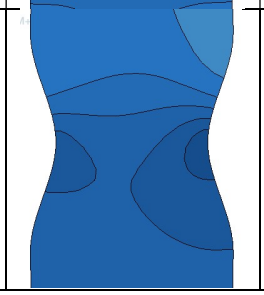
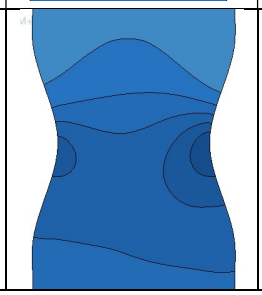
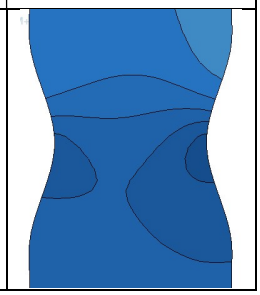
1st order				
2nd order				

Table 6. Velocity vector plots in the stenosis region, comparing different mesh types with 1st- and 2nd-order discretization.

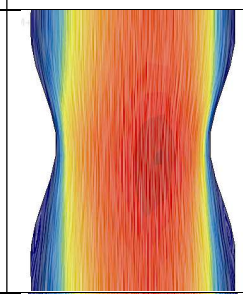
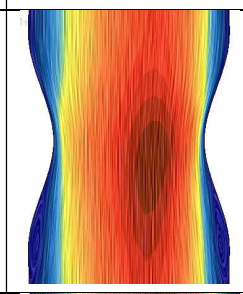
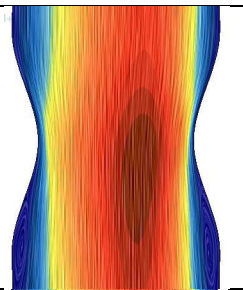
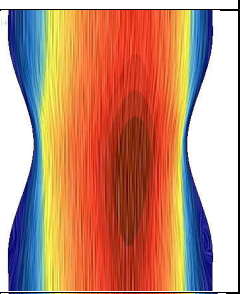
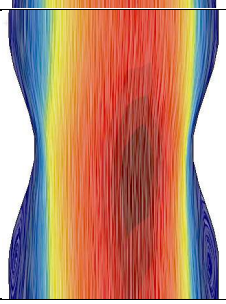
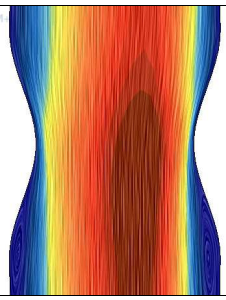
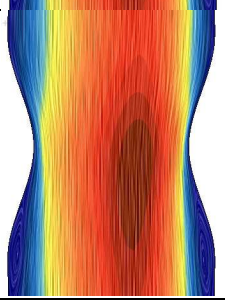
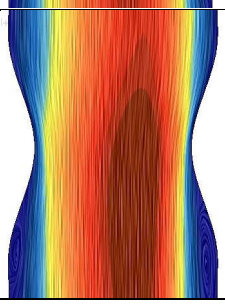
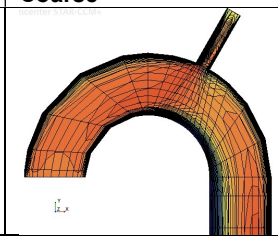
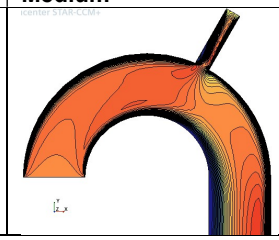
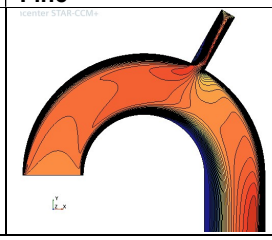
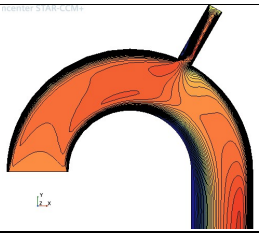
Type of mesh:	Coarse	Medium	Fine	Other
1st order				
2nd order				

Table 7. Contour plots of velocity magnitude in the small branch region, comparing various mesh types with 1st- and 2nd-order discretization.

	Coarse	Medium	Fine	Other
1st order				

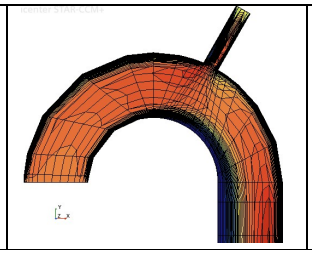
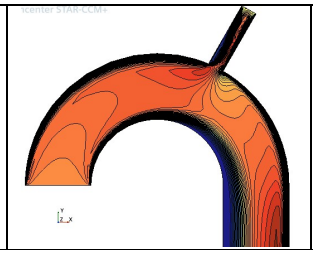
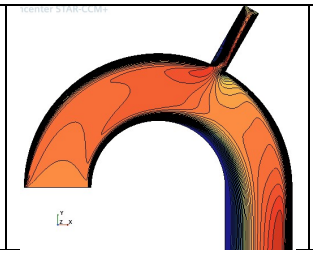
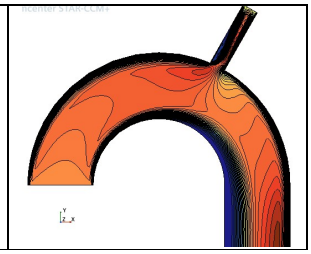
2nd order				

Table 8. Contour plots of pressure magnitude in the small branch region, comparing various mesh types with 1st- and 2nd-order discretization.

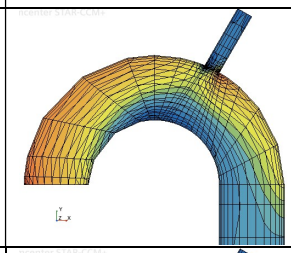
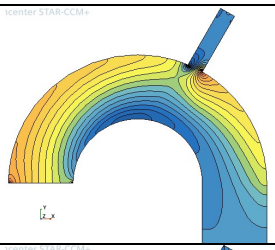
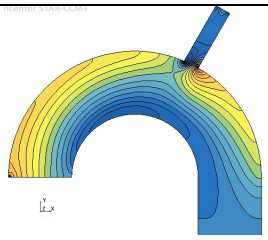
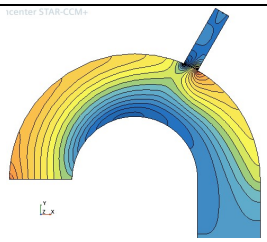
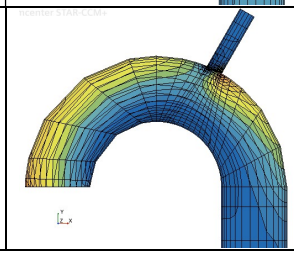
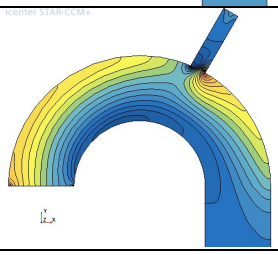
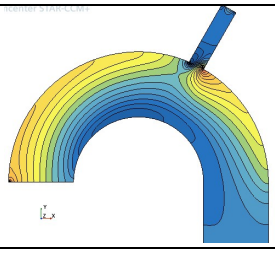
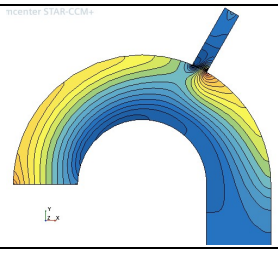
Type of mesh:	Coarse	Medium	Fine	Other
1st order				
2nd order				

Table 9. Velocity vector plots in the small branch region, comparing different mesh types with 1st- and 2nd-order discretization.

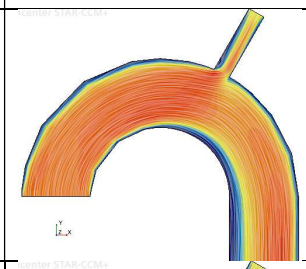
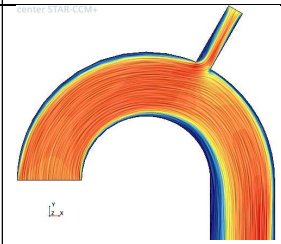
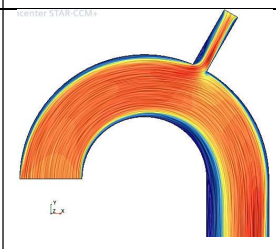
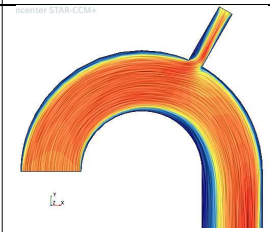
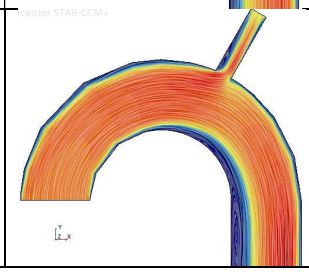
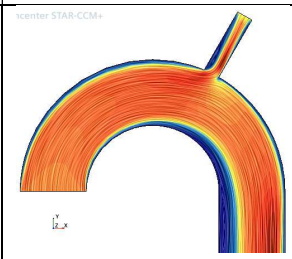
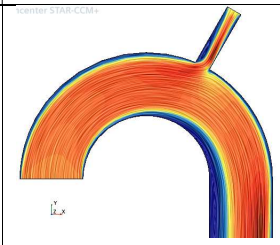
Type of mesh:	Coarse	Medium	Fine	Other
1st order				
2nd order				

Table 10. Contour plots of velocity magnitude in the Subclavian artery inlet, comparing various mesh types with 1st- and 2nd-order discretization.

Type	Coarse	Medium	Fine	Other

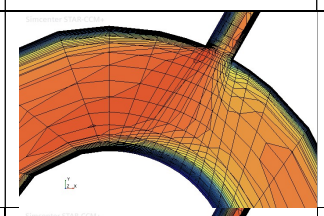
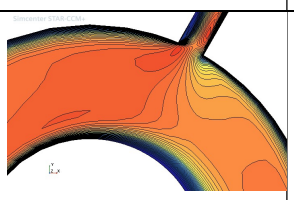
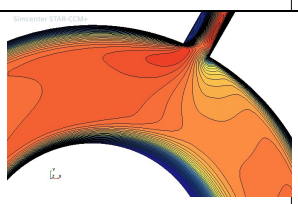
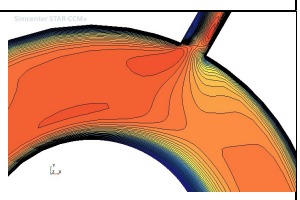
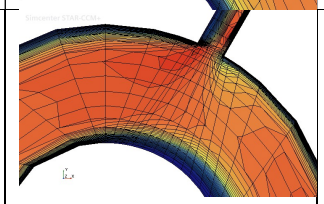
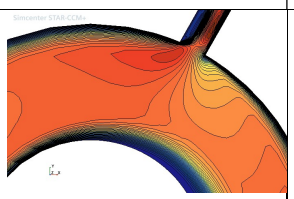
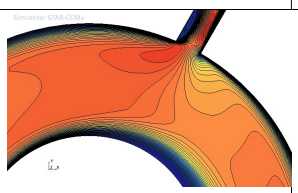
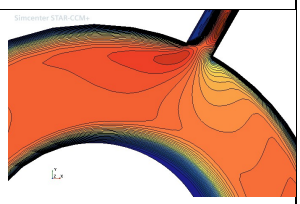
of mesh:				
1st order				
2nd order				

Table 11. Contour plots of pressure magnitude in the Subclavian artery inlet, comparing various mesh types with 1st- and 2nd-order discretization.

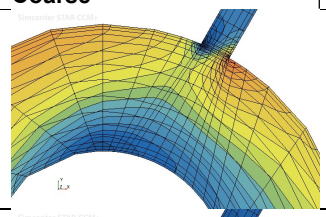
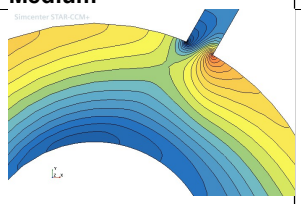
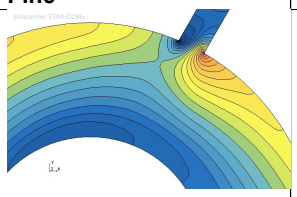
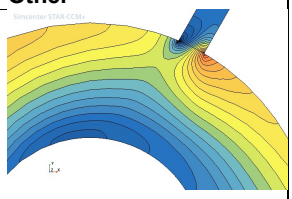
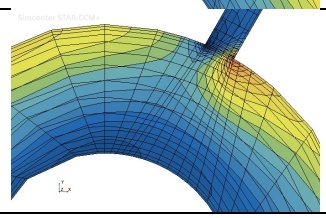
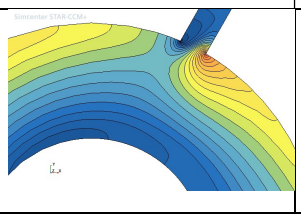
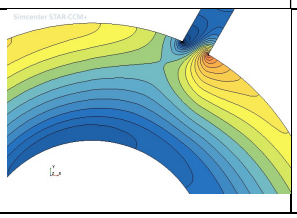
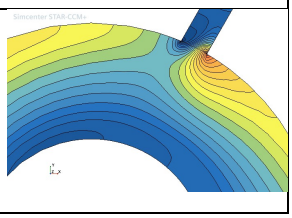
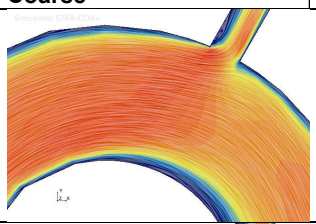
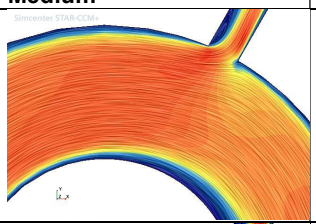
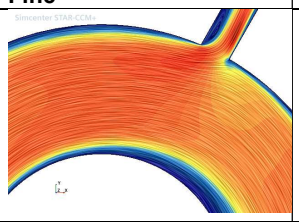
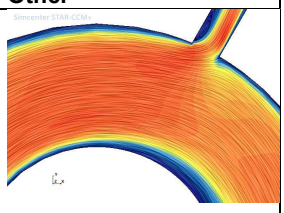
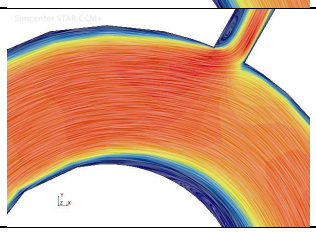
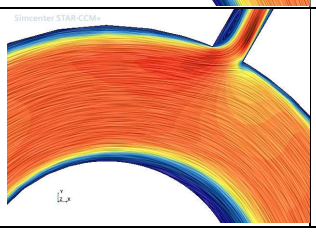
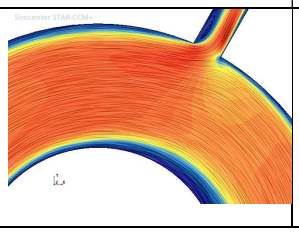
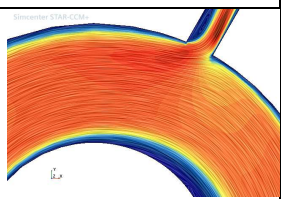
	Coarse	Medium	Fine	Other
1st order				
2nd order				

Table 12. Velocity vector plots at the Subclavian artery inlet, comparing different mesh types with 1st- and 2nd-order discretization.

	Coarse	Medium	Fine	Other
1st order				
2nd order				

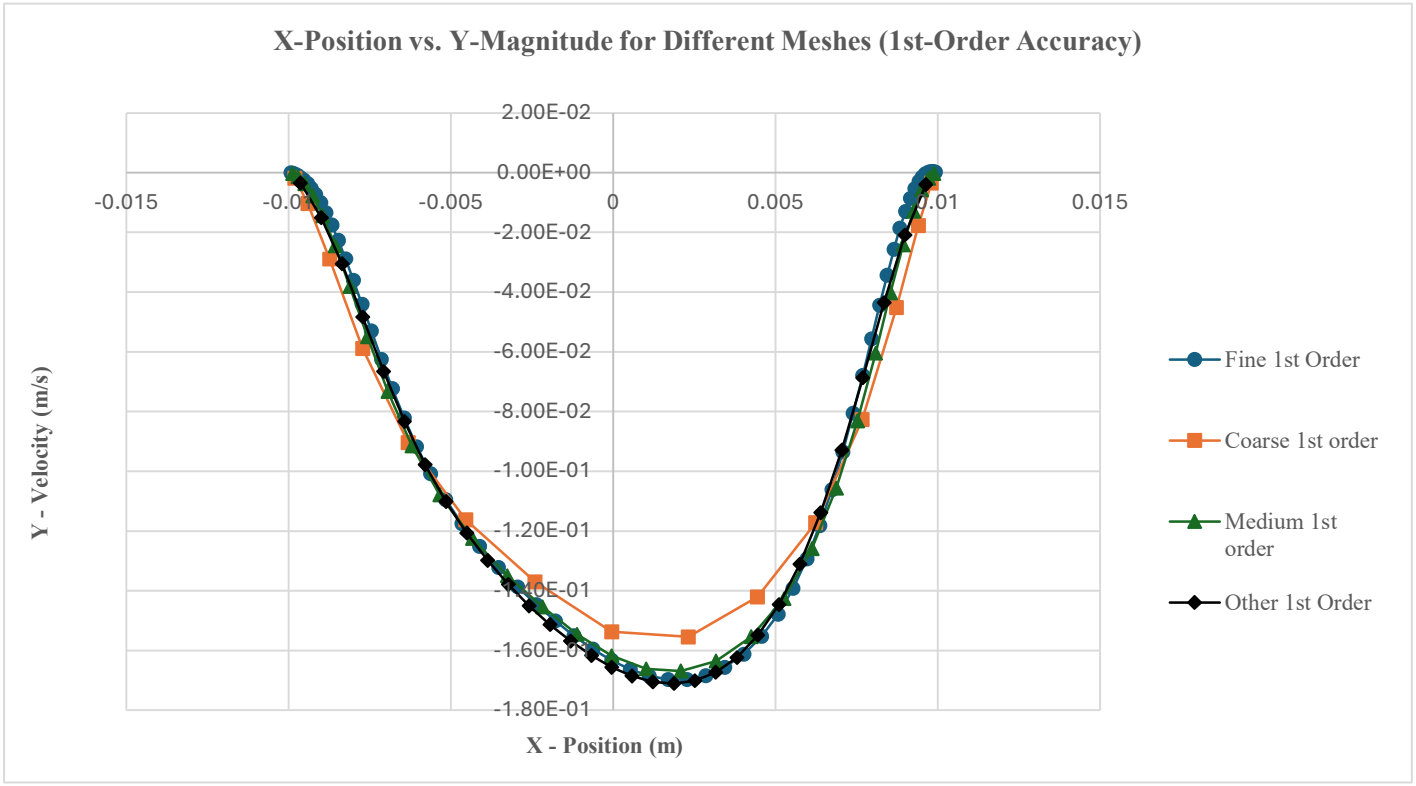


Figure 2 – Combined XY plots of position vs. magnitude for different meshes using 1st-order discretization.

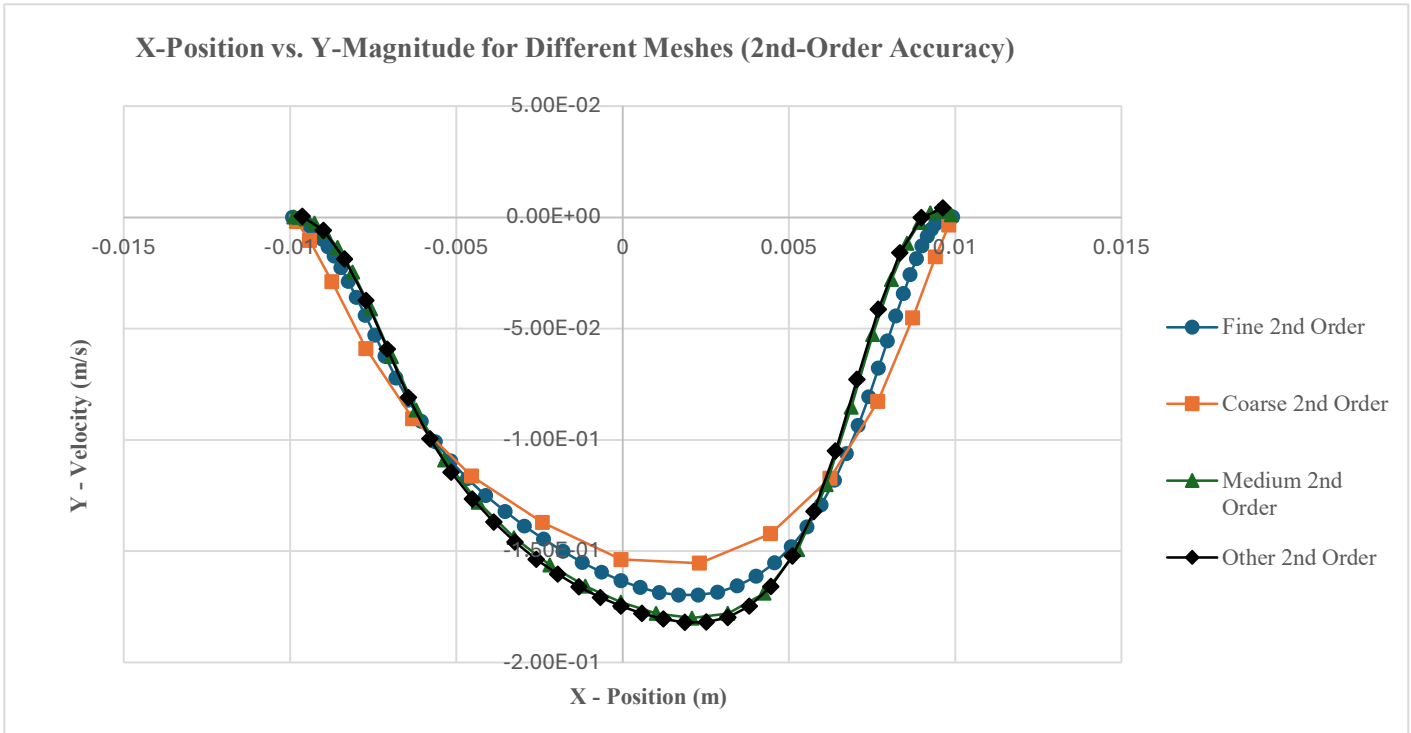


Figure 3 – Combined XY plots of position vs. magnitude for different meshes using 2nd-order discretization

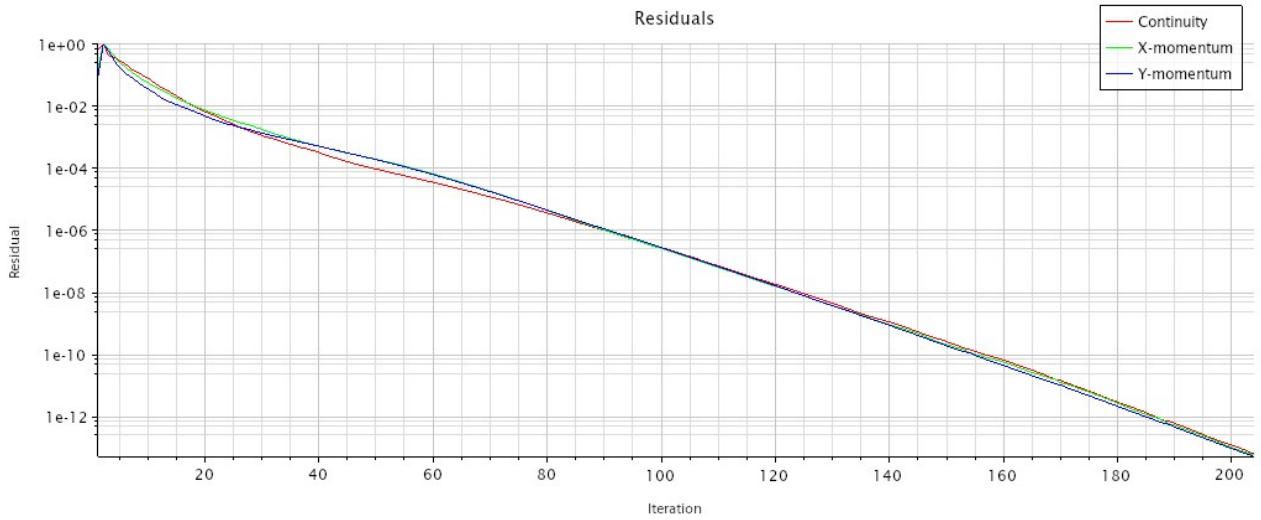


Figure 4 - Residuals convergence graph for coarse mesh using 1st-order discretization

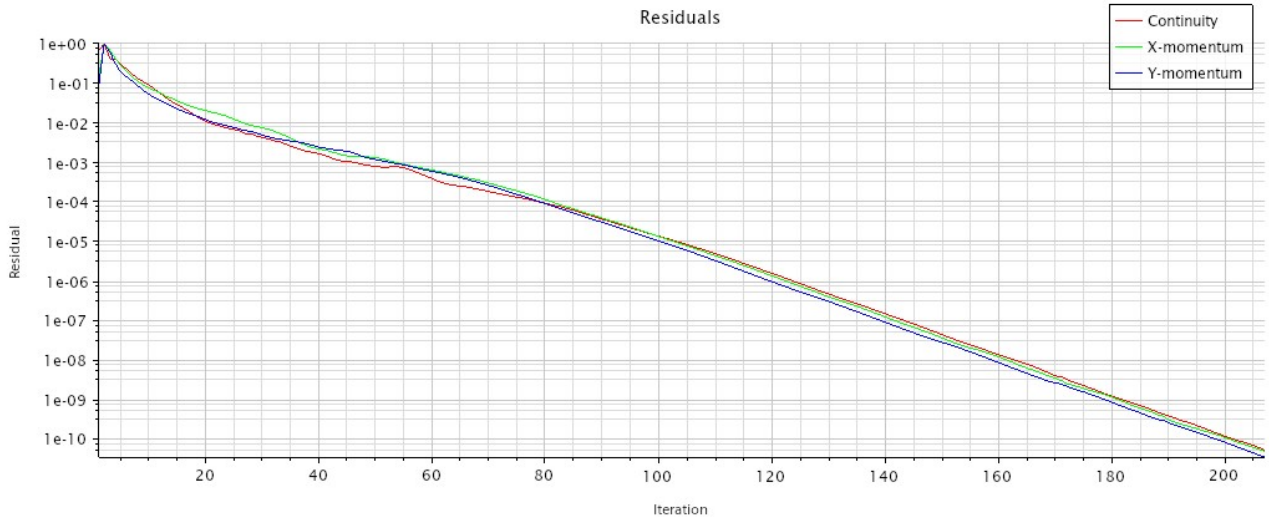


Figure 5 - Residuals convergence graph for coarse mesh using 2nd-order discretization.

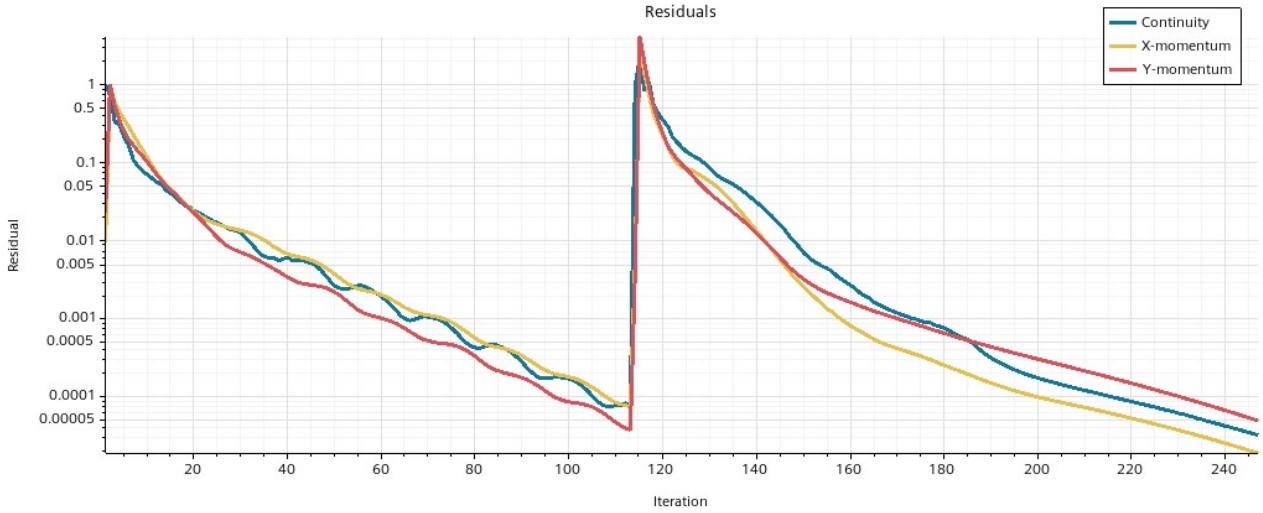


Figure 6 - Residuals convergence graph for medium mesh using 1st-order discretization.

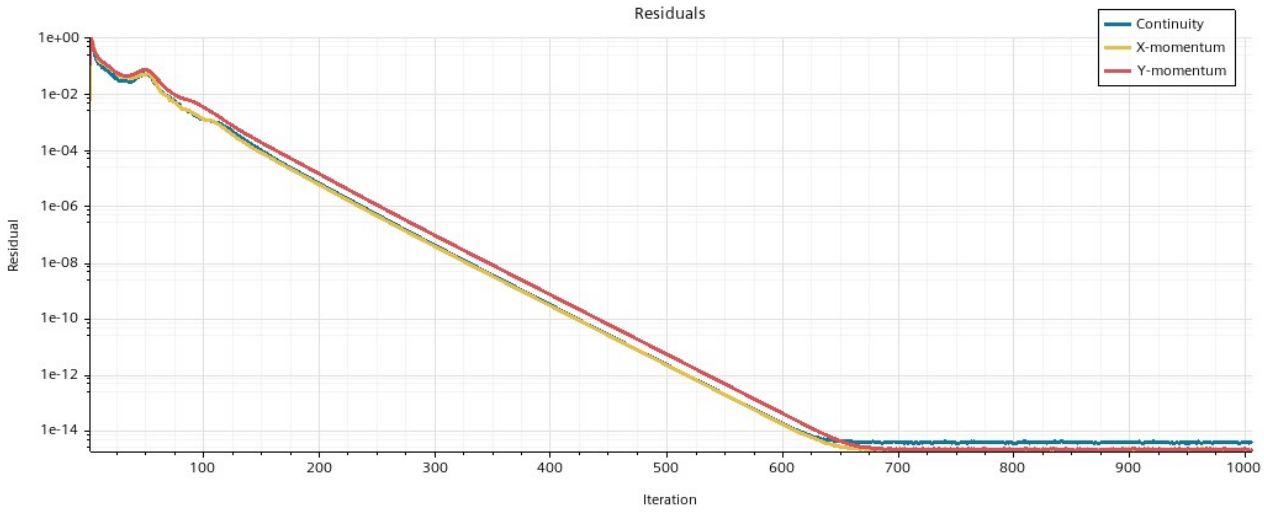


Figure 7 - Residuals convergence graph for medium mesh using 2nd-order discretization.

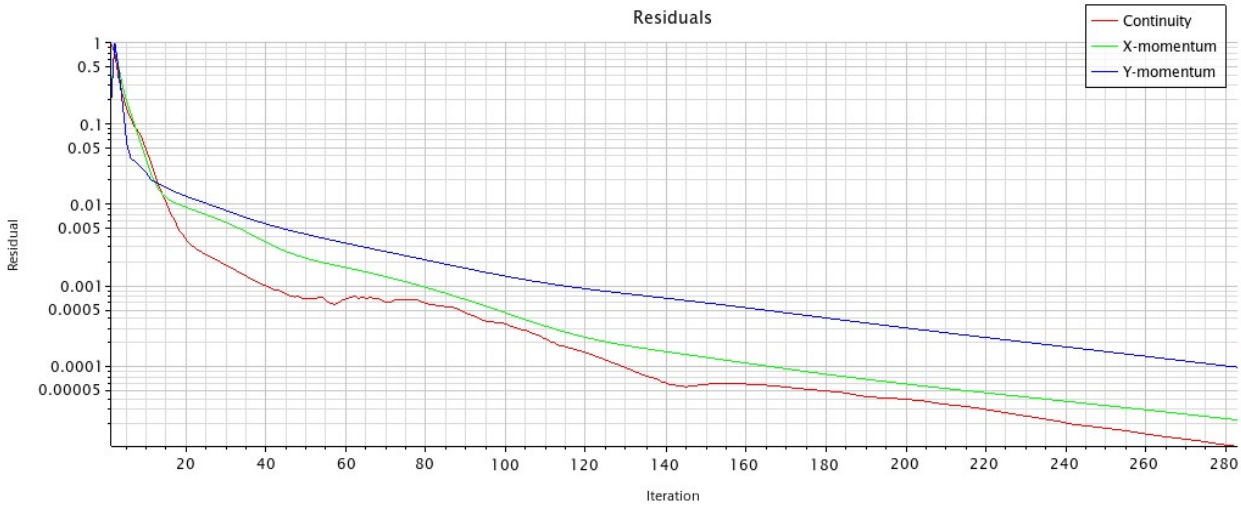


Figure 8 - Residuals convergence graph for fine mesh using 1st-order discretization.

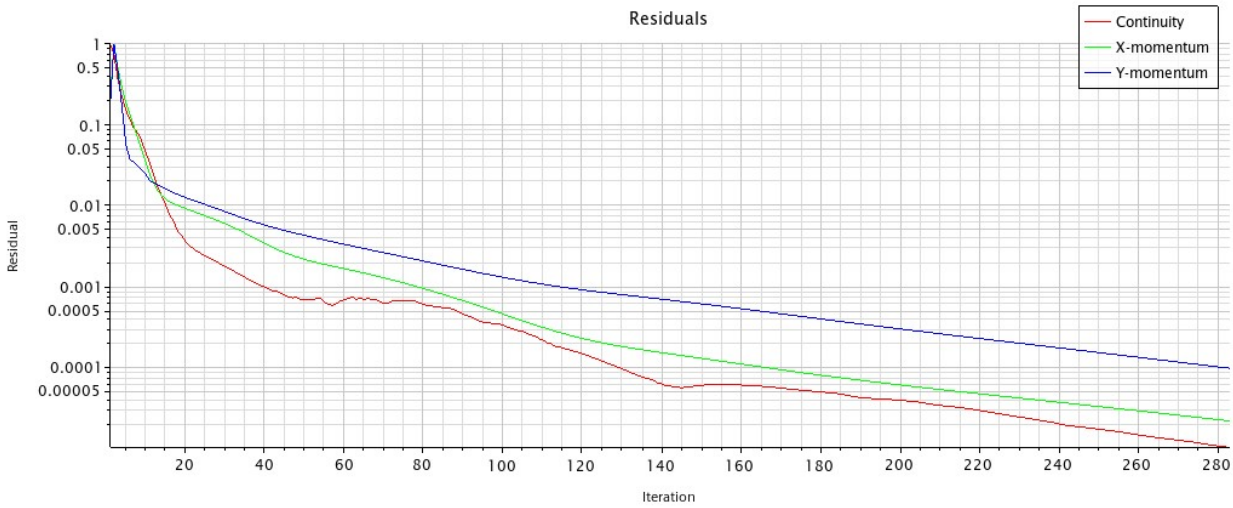


Figure 9 - Residuals convergence graph for fine mesh using 2nd-order discretization.

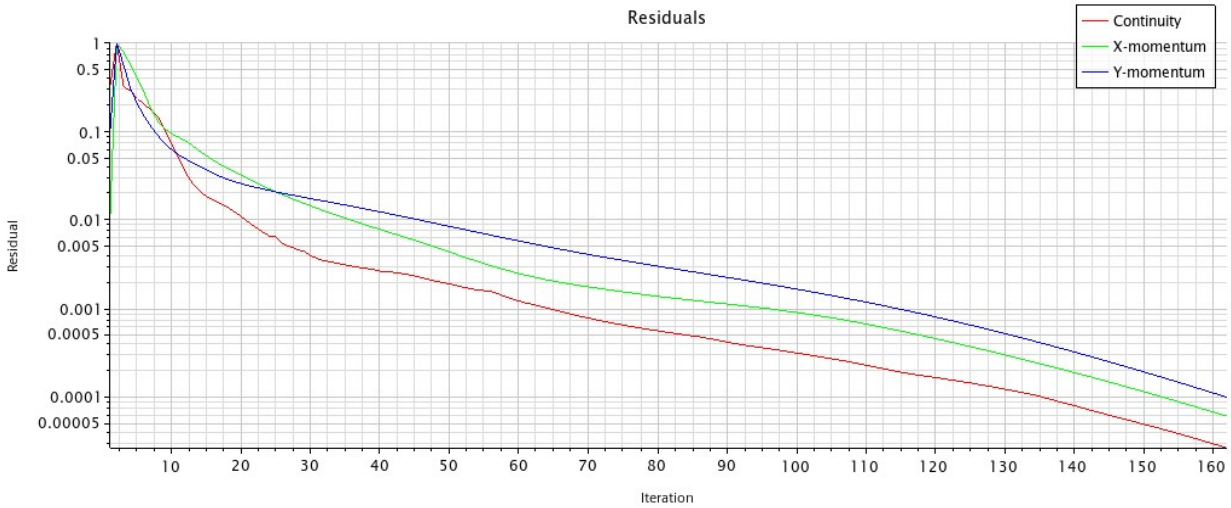


Figure 10 - Residuals convergence graph for other mesh using 1st-order discretization.

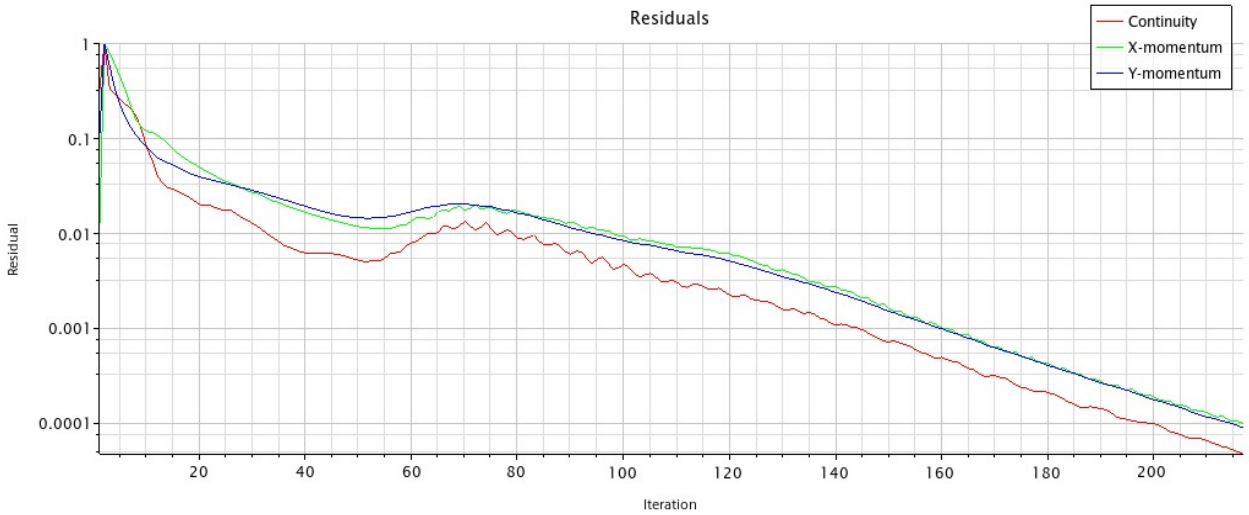


Figure 11 - Residuals convergence graph for 'Other' mesh using 2nd-order discretization.